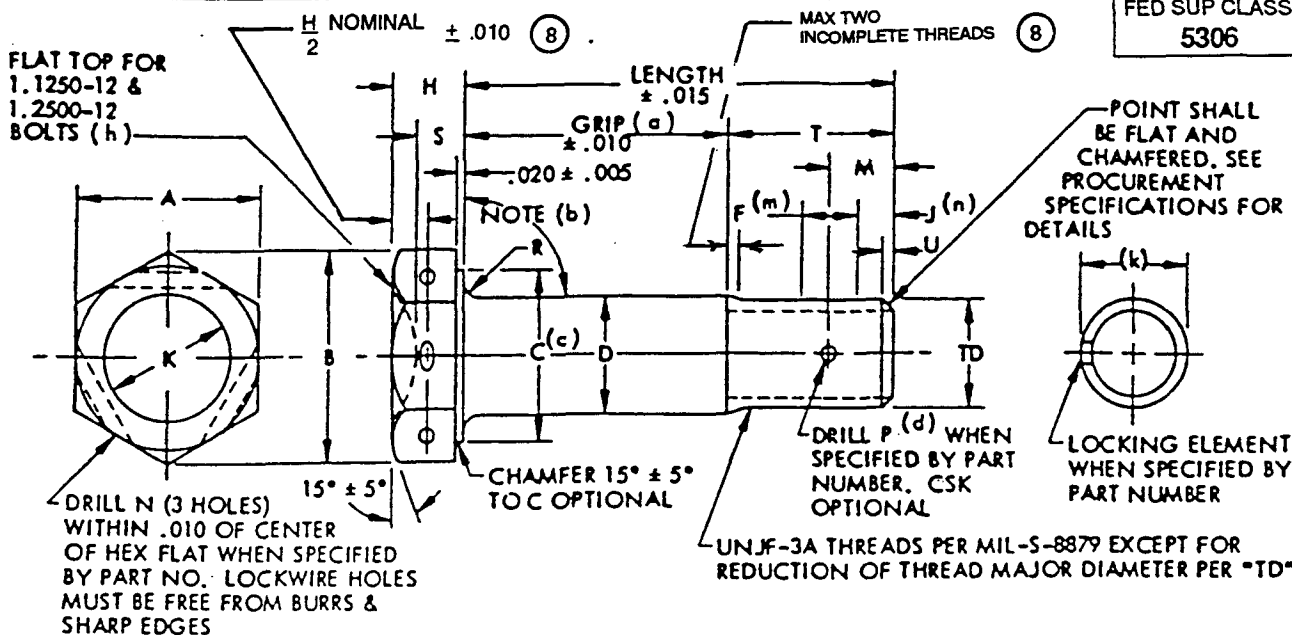


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HEAD MARKING: Basic number plus grip dash number plus "D,"
"L" or "P", when applicable, plus manufacturer's symbol,
raised or depressed .010 MAX. Arrangement optional.
"D" identifies bolt with drilled shank.
"L" identifies bolt with optional locking element.
"P" identifies bolt with patch type locking element only.
"A," aluminum coated and "U," unplated codes need not appear on bolt head.
"E" or "G" identifies chromium plated bolts with shot peening.

TABLE I DIMENSIONS (CONTINUED ON SHEET 2)

BASIC NUMBER **	THREAD UNJF-3A	A		B MIN	C DIA (c)	D DIA UNPLATED	
		MAX	MIN			MAX	MIN
NAS6703	.1900-32	.376	.367	.410	.335	.1895	.1890
NAS6704	.2500-28	.439	.429	.480	.398	.2495	.2490
NAS6705	.3125-24	.502	.492	.552	.460	.3120	.3115
NAS6706	.3750-24	.564	.554	.623	.523	.3745	.3740
NAS6707	.4375-20	.690	.678	.764	.648	.4370	.4365
NAS6708	.5000-20	.752	.741	.836	.710	.4995	.4990
NAS6709	.5625-18	.877	.865	.978	.835	.5615	.5610
NAS6710	.6250-18	.940	.928	1.050	.898	.6240	.6235
NAS6712	.7500-16	1.065	1.052	1.191	1.023	.7490	.7485
NAS6714	.8750-14	1.252	1.239	1.405	1.210	.8740	.8735
NAS6716	1.0000-12	1.440	1.427	1.619	1.398	.9990	.9985
NAS6718	1.1250-12	1.627	1.614	1.832	1.585	1.1240	1.1230
NAS6720	1.2500-12	1.814	1.801	2.046	1.772	1.2490	1.2480

** The last two digits of the basic number designate the size.
See sheet 3 for "D", "H", "L", "P" and "U" codes.
See sheets 4 and 5 for grip dash numbers.
(a), (b) etc. See notes on sheet 4.

LIST OF CURRENT SHEETS

SHT.	REV.	SHT.	REV.
1	8	5	3
2	4	6	2
3	6	7	2
4	4		

CUSTODIAN NATIONAL AEROSPACE STANDARDS COMMITTEE		THIRD ANGLE PROJECTION
PROCUREMENT SPECIFICATION NOTED ON SHEET 3	TITLE BOLT, HEX HEAD, CLOSE TOLERANCE, A286 CRES, LONG THREADS, SELF-LOCKING AND NONLOCKING	CLASSIFICATION STANDARD PART
		NAS6703 THRU 6720
		SHEET 1 OF 7

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TABLE I DIMENSIONS (CONTINUED)

BASIC NUMBER	D DIAMETER, PLATED OR COATED (r) (4)								F (m)
	BEFORE CHROME PLATE		BEFORE CAD PLATE		BEFORE AL COATING		AFTER PLATING OR COATING		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
NAS6703	.1850	.1845	.1885	.1879	.1887	.1881	.1895	.1885	.156
NAS6704	.2450	.2445	.2485	.2479	.2487	.2481	.2495	.2485	.178
NAS6705	.3075	.3070	.3110	.3104	.3112	.3106	.3120	.3110	.208
NAS6706	.3700	.3695	.3735	.3729	.3737	.3731	.3745	.3735	.208
NAS6707	.4325	.4320	.4360	.4354	.4362	.4356	.4370	.4360	.250
NAS6708	.4950	.4945	.4985	.4979	.4987	.4981	.4995	.4985	.250
NAS6709	.5570	.5565	.5605	.5599	.5607	.5601	.5615	.5605	.278
NAS6710	.6195	.6190	.6230	.6224	.6232	.6226	.6240	.6230	.278
NAS6712	.7445	.7440	.7480	.7474	.7482	.7476	.7490	.7480	.312
NAS6714	.8695	.8690	.8730	.8724	.8732	.8726	.8740	.8730	.357
NAS6716	.9945	.9940	.9980	.9974	.9982	.9976	.9990	.9980	.416
NAS6718	1.1195	1.1185	1.1230	1.1219	1.1232	1.1221	1.1240	1.1225	.416
NAS6720	1.2445	1.2435	1.2480	1.2469	1.2482	1.2471	1.2490	1.2475	.416

(4)

(4)

(4)

TABLE I DIMENSIONS (CONTINUED)

BASIC NUMBER	H +.015 -.000	J (n)	K DIA ±.010	M ±.010	N DIA +.010 -.000	P (d) DIA +.010 -.000	R RAD		S +.015 -.000	T (e) REF
							MAX	MIN		
NAS6703	.110	.094	.19	.164	.046	.070	.020	.010	.073	.345
NAS6704	.125	.107	.25	.178	.046	.076	.020	.010	.083	.425
NAS6705	.156	.125	.31	.181	.070	.076	.020	.010	.104	.469
NAS6706	.188	.125	.38	.197	.070	.106	.025	.015	.125	.578
NAS6707	.219	.150	.44	.201	.070	.106	.025	.015	.146	.694
NAS6708	.250	.150	.50	.216	.070	.106	.030	.020	.167	.735
NAS6709	.281	.167	.56	.218	.070	.141	.035	.020	.188	.840
NAS6710	.312	.167	.62	.249	.070	.141	.040	.025	.208	.902
NAS6712	.375	.188	.75	.252	.070	.141	.045	.030	.250	1.041
NAS6714	.438	.214	.88	.257	.070	.141	.050	.035	.292	1.184
NAS6716	.500	.250	1.00	.264	.070	.141	.060	.045	.333	1.309
NAS6718	.562	.250	(h)	.357	.070	.141	.070	.055	(h)	1.458
NAS6720	.625	.250	(h)	.389	.070	.141	.075	.060	(h)	1.646

TABLE I DIMENSIONS (CONTINUED)

NOM SIZE	U MAX	TD DIA		INSPECTION DATA		
		MAX	MIN	X ^(f)	Y ^{(f)(g)}	Z ^(g)
.1900	.039	.184	.181	.005	.0045	.0040
.2500	.045	.244	.241	.006	.0045	.0030
.3125	.052	.306	.302	.008	.0045	.0030
.3750	.052	.368	.364	.009	.0045	.0025
.4375	.062	.431	.426	.010	.006	.0025
.5000	.062	.493	.488	.011	.006	.0020
.5625	.068	.555	.550	.012	.006	.0020
.6250	.068	.618	.612	.015	.006	.0020
.7500	.078	.743	.737	.018	.009	.0020
.8750	.089	.868	.861	.020	.009	.0020
1.0000	.104	.993	.986	.022	.009	.0020
1.1250	.104	1.118	1.111	.025	.009	.0020
1.2500	.104	1.243	1.236	.028	.009	.0020

(d), (e), ETC: See Notes on Sheet 3.

NAS6703 THRU 6720
SHEET 2

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APPROVAL DATE FEB. 1969 REVISION (1) 16 MARCH 1970 (2) 15 JUNE 1976 (3) 17 DEC. 1993 (4) 25 FEBRUARY 1994



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- MATERIAL:** CRES - A286 (UNS S66286) conforming to the chemistry of AMS5731, AMS5732, AMS5737 or AMS5853. (6)
- HEAT TREAT:** Locking element - Plastic per MIL-F-18240 and QPL18240.
Develop basic material properties as follows, with controls per MIL-H-6875. (6)
160-190 KSI Ft_u, 95 KSI minimum ultimate shear. Precipitation hardening per AMS5853. (1)
- FINISH:** Unplated bolts - Passivate per QQ-P-35.
Cadmium Plated bolts - Cadmium plate per QQ-P-416, Type II, Class 2. ~~Parts plated to Class 3 may be used until stock depleted.~~ Post-plate hydrogen embrittlement baking and testing per QQ-P-416 (6)
are not required for A286 material.
- Chromium Plated bolts**
- Code C - Chromium plate per QQ-C-320, class 2 on shank only. All other surfaces cadmium plated. No chromium within .020 of line of tangency of head-to-shank fillet. Chromium in thread runout permitted. Chromium plated bolts not available with grip dash number 1 or number 2.
- Code E - Chromium plate per QQ-C-320, class 2a. Plating on shank only. No plating on: Top or sides of head, bearing surface of head, head to shank fillet, adjacent to fillet (within .020 minimum to .080 maximum of line of tangency between shank and radius R), and threads except runout may be plated. All surfaces not chromium plated shall be cadmium plated per QQ-P-416, Type II, Class 2.
- Bolts shall be shot peened per MIL-S-13165 and the following requirements: Cast steel shot of size 170 to 390, intensity .014A to .018A, shot peen prior to chromium plating, thread rolling and cold working "R" radius, but after heat treating and grinding. Minimum areas to be shot peened: bearing surface of head, head to shank fillet, shank, and thread runout.
- Chromium plated bolts not available with grip dash number 1 or number 2.
- Code G - All requirements identical to Code E except delete requirement for cadmium plating of non-chrome plated surfaces and add the following requirement: Passivate entire bolt per QQ-P-35 after chromium plating.
- Coated bolts - Aluminum coating per NAS4006.
- Cadmium plated A-286 CRES bolts shall be identified with green dye or paint on the thread end. Maximum coverage may include the chamfer plus one incomplete thread.
- CODE:** Basic part number = nonlocking bolt with undrilled shank and undrilled head, cadmium plated. Add "A" after basic number for aluminum coated bolts. May be used with "D", "H", "L" or "P" code.
- (6) Add "D" after basic number for drilled shank bolts. Do not use with "L" or "P" code. (U)
- (6) Add "H" after basic number for bolt with drilled head. ~~Do not use with "L" or "P" code.~~
- Add "L" after basic number for self-locking bolt with optional locking element; see procurement spec below. Do not use with "D", "H" or "P" code.
- Add "P" after basic number for self-locking bolt with patch type locking element only; see procurement spec below. Do not use with "D", "H" or "L" code.
- Add "U" after basic number for unplated bolt. May be used with "D", "H", "L" or "P" code.
- Add "C" after basic number for chromium plated bolts not requiring shot peening. Surfaces not chromium plated are cadmium plated. Do not use with "A", "E", "U", or "G" codes.
- Add "E" after basic number for chromium plated bolts with mandatory shot peening. Surfaces not chromium plated are cadmium plated. Do not use with "A", "C", "U" or "G" codes.
- Add "G" after basic number for chromium plated bolts with mandatory shot peening. Surfaces not chromium plated are passivated only. Do not use with "A", "C", "U" or "E" codes.
- When multiple letter codes are used, sequence must be in alphabetical order. See Sheet 4 and 5 for tabulations of grip and length dimensions.

NAS6703 THRU 6720

SHEET 3

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EXAMPLES OF PART NUMBERS: (See Sheet 7 for oversize bolts).

NAS6704-10	= Bolt, .2500-28 thread, .625 grip, undrilled, nonlocking, cadmium plated.
NAS6704A10	= Bolt, .2500-28 thread, .625 grip, undrilled, nonlocking, aluminum coated
NAS6704D10	= Bolt, .2500-28 thread, .625 grip, drilled shank, undrilled head, nonlocking, cadmium plated.
NAS6704H10	= Bolt, .2500-28 thread, .625 grip, drilled head, undrilled shank, nonlocking, cadmium plated.
NAS6704DH10	= Bolt, .2500-28 thread, .625 grip, drilled shank, drilled head, nonlocking, cadmium plated.
NAS6704L10	= Bolt, .2500-28 thread, .625 grip, undrilled, self-locking (optional type; see procurement spec below), cadmium plated.
NAS6704P10	= Bolt, .2500-28 thread, .625 grip, undrilled, self-locking (patch type; see procurement spec below), cadmium plated.
NAS6704PU10	= Bolt, .2500-28 thread, .625 grip, undrilled, self-locking (patch type), unplated.
NAS6704C10	= Bolt, .2500-28 thread, .625 grip, chromium plated, without mandatory shot peening. Surfaces not chromium plated are cadmium plated.
NAS6704E10	= Bolt, .2500-28 thread, .625 grip, chromium plated, with mandatory shot peening, surfaces not chrome plated are cad plated.
NAS6704G10	= Bolt, .2500-28 thread, .625 grip, chromium plated, with mandatory shot peening, surfaces not chrome plated are passivated.

- NOTES: (a) Grip length: From under side of head to end of full cylindrical portion of shank.
- (b) Bearing surface squareness: Within .003 FIM with shank.
- (c) Washer face diameter: Max not to exceed actual width across flats; min as tabulated in Table I.
- (d) Cotter pin hole centerline: Within .010 and normal within 2° of bolt centerline.
- (e) Reference dimensions are for design purposes only and are not an inspection requirement.
- (f) Concentricity: "C" and "D" diameters within "X" values FIM. "D" and thread pitch dia within "Y" values FIM.
- (g) Shank straightness: Within "Z" values FIM per inch of length.
- (h) Top of head shall be flat on the 1.1250-12 and 1.2500-12 size bolts.
- (k) Protrusion of locking element shall be controlled so that it will pass freely, or with finger pressure, through a ring gage with diameter of .010 (+.001, -.000) greater than maximum major diameter of bolt thread.
- (m) "F" min (5 thread pitches) = region of minimum engagement with female thread required to meet MIL-F-18240 requirements. Locking element within "F" region must develop required torque when tested per MIL-F-18240.
- (n) For ease in starting, locking element shall not be effective in "J" area (3 thread pitches).
- (p) Magnetic permeability shall be less than 2.0 (Air = 1.0) for a field strength H = 200 oersteds using a magnetic permeability indicator per ASTM A342.
- (q) For chromium plated parts, the cylinder described by the thread major diameter (dimension "TD DIA") must fall within the cylinder described by the shank diameter (dimension "D DIAMETER, AFTER PLATING OR COATING").
- (r) ~~Plating thickness minimum to be .0003 inch per QQ-P-416, Class 2~~

NOTES CONTINUED ON SHEET 5.

SURFACE TEXTURE: (AA max per ANSI/ASME B46.1) "D" dia., head-to-shank fillet, bearing surface of head, thread flanks and thread root 32; other surfaces 125.

Dimensions to be met after plating or coating, except as noted on Sheets 1 and 7.
Dimensions in inches.

PROCUREMENT SPECIFICATION: NAS4003, except as noted. Cold work of head to shank fillet radius is not required for NAS6703 bolts. Locking element for self-locking bolts; Per MS15981 and MIL-F-18240, any type or configuration, including patch type, is optional when "L" code is specified. Patch type locking element (with no metal removed) is required when "P" code is specified. Locking element must be supplied by a qualified source listed in QPL18240 or approved for listing in QPL18240. Shipping notice should identify supplier of bolt and locking element separately.

**NAS6703 THRU 6720
SHEET 4**

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TABLE II GRIP AND LENGTH DIMENSIONS (CONTINUED ON SHEET 6)

GRIP DASH NO.	GRIP ± .010	LENGTHS ±.015 (SEE NOTES BELOW)						
		BASIC NUMBER AND THREAD SIZE						
		NAS6703 .1900-32	NAS6704 .2500-28	NAS6705 .3125-24	NAS6706 .3750-24	NAS6707 .4375-20	NAS6708 .5000-20	NAS6709 .5625-18
1	.062	.407	.487	.531	.640	.756	.797	.902
2	.125	.470	.550	.594	.703	.819	.860	.965
3	.188	.533	.613	.657	.766	.882	.923	1.028
4	.250	.595	.675	.719	.828	.944	.985	1.090
5	.312	.657	.737	.781	.890	1.006	1.047	1.152
6	.375	.720	.800	.844	.953	1.069	1.110	1.215
7	.438	.783	.863	.907	1.016	1.132	1.173	1.278
8	.500	.845	.925	.969	1.078	1.194	1.235	1.340
9	.562	.907	.987	1.031	1.140	1.256	1.297	1.402
10	.625	.970	1.050	1.094	1.203	1.319	1.360	1.465
11	.688	1.033	1.113	1.157	1.266	1.382	1.423	1.528
12	.750	1.095	1.175	1.219	1.328	1.444	1.485	1.590
13	.812	1.157	1.237	1.281	1.390	1.506	1.547	1.652
14	.875	1.220	1.300	1.344	1.453	1.569	1.610	1.715
15	.938	1.283	1.363	1.407	1.516	1.632	1.673	1.778
16	1.000	1.345	1.425	1.469	1.578	1.694	1.735	1.840
17	1.062	1.407	1.487	1.531	1.640	1.756	1.797	1.902
18	1.125	1.470	1.550	1.594	1.703	1.819	1.860	1.965
19	1.188	1.533	1.613	1.657	1.766	1.882	1.923	2.028
20	1.250	1.595	1.675	1.719	1.828	1.944	1.985	2.090
21	1.312	1.657	1.737	1.781	1.890	2.006	2.047	2.152
22	1.375	1.720	1.800	1.844	1.953	2.069	2.110	2.215
23	1.438	1.783	1.863	1.907	2.016	2.132	2.173	2.278
24	1.500	1.845	1.925	1.969	2.078	2.194	2.235	2.340
25	1.562	1.907	1.987	2.031	2.140	2.256	2.297	2.402
26	1.625	1.970	2.050	2.094	2.203	2.319	2.360	2.465
27	1.688	2.033	2.113	2.157	2.266	2.382	2.423	2.528
28	1.750	2.095	2.175	2.219	2.328	2.444	2.485	2.590
29	1.812	2.157	2.237	2.281	2.390	2.506	2.547	2.652
30	1.875	2.220	2.300	2.344	2.453	2.569	2.610	2.715
31	1.938	2.283	2.363	2.407	2.516	2.632	2.673	2.778
32	2.000	2.345	2.425	2.469	2.578	2.694	2.735	2.840
34	2.125	2.470	2.550	2.594	2.703	2.819	2.860	2.965
36	2.250	2.595	2.675	2.719	2.828	2.944	2.985	3.090
38	2.375	2.720	2.800	2.844	2.953	3.069	3.110	3.215
40	2.500	2.845	2.925	2.969	3.078	3.194	3.235	3.340
42	2.625	2.970	3.050	3.094	3.203	3.319	3.360	3.465
44	2.750	3.095	3.175	3.219	3.328	3.444	3.485	3.590
46	2.875	3.220	3.300	3.344	3.453	3.569	3.610	3.715
48	3.000	3.345	3.425	3.469	3.578	3.694	3.735	3.840
50	3.125	3.470	3.550	3.594	3.703	3.819	3.860	3.965
52	3.250	3.595	3.675	3.719	3.828	3.944	3.985	4.090
54	3.375	3.720	3.800	3.844	3.953	4.069	4.110	4.215
56	3.500	3.845	3.925	3.969	4.078	4.194	4.235	4.340
58	3.625	3.970	4.050	4.094	4.203	4.319	4.360	4.465
60	3.750	4.095	4.175	4.219	4.328	4.444	4.485	4.590
62	3.875	4.220	4.300	4.344	4.453	4.569	4.610	4.715
64	4.000	4.345	4.425	4.469	4.578	4.694	4.735	4.840

INTERMEDIATE AND LONGER LENGTHS MAY BE SPECIFIED BY USE OF WHOLE GRIP DASH NUMBERS ONLY.
 NOMINAL GRIP DIMENSION EQUALS GRIP DASH NUMBER TIMES .0625 (ROUNDED TO 3 DECIMAL PLACES).
 NOMINAL LENGTH EQUALS NOMINAL GRIP PLUS "T" (SEE SHEET 2).

NOTES CONTINUED FROM SHEET 4.

(3) (t)

The effect of cold work and aging induced during the manufacturing cycle may increase the ultimate tensile value of the finished part, but this shall not exceed 1.3 times the specified minimum tensile value.

(3) (u)

If required, tensile testing of bolts requiring cross-drilled threads shall be performed prior to drilling and the application of plating and/or coatings. When bolts have been drilled, strength may be verified by shear testing, in lieu of tensile testing, in accordance with MIL-STD-1312. Users should be aware that fasteners with cross-drilled threads may exhibit a reduction in tensile strength.

NAS 6703 THRU 6720
SHEET 5

NATIONAL AEROSPACE STANDARD

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC., 1725 DE SALES STREET, N.W. WASHINGTON, D. C. 20036

TABLE II GRIP AND LENGTH DIMENSIONS (CONTINUED)

GRIP DASH NO.	GRIP ±.010	LENGTH ±.015 (SEE NOTES BELOW)					
		BASIC NUMBER AND THREAD SIZE					
		NAS6710 .6250-18	NAS6712 .7500-16	NAS6714 .8750-14	NAS6716 1.0000-12	NAS6718 1.1250-12	NAS6720 1.2500-12
1	.062	.964	1.103	1.246	1.371	1.520	1.708
2	.125	1.027	1.166	1.309	1.434	1.583	1.771
3	.188	1.090	1.229	1.372	1.497	1.646	1.834
4	.250	1.152	1.291	1.434	1.559	1.708	1.896
5	.312	1.214	1.353	1.496	1.621	1.770	1.958
6	.375	1.277	1.416	1.559	1.684	1.883	2.021
7	.438	1.340	1.479	1.622	1.747	1.896	2.084
8	.500	1.402	1.541	1.684	1.809	1.958	2.146
9	.562	1.464	1.603	1.746	1.871	2.020	2.208
10	.625	1.527	1.666	1.809	1.934	2.083	2.271
11	.688	1.590	1.729	1.872	1.997	2.146	2.334
12	.750	1.652	1.791	1.934	2.059	2.208	2.396
13	.812	1.714	1.853	1.996	2.121	2.270	2.458
14	.875	1.777	1.916	2.059	2.184	2.333	2.521
15	.938	1.840	1.979	2.122	2.247	2.396	2.584
16	1.000	1.902	2.041	2.184	2.309	2.458	2.646
17	1.062	1.964	2.103	2.246	2.371	2.520	2.708
18	1.125	2.027	2.166	2.309	2.434	2.583	2.771
19	1.188	2.090	2.229	2.372	2.497	2.646	2.834
20	1.250	2.152	2.291	2.434	2.559	2.708	2.896
21	1.312	2.214	2.353	2.496	2.621	2.770	2.958
22	1.375	2.277	2.416	2.559	2.684	2.833	3.021
23	1.438	2.340	2.479	2.622	2.747	2.896	3.084
24	1.500	2.402	2.541	2.684	2.809	2.958	3.146
25	1.562	2.464	2.603	2.746	2.871	3.020	3.208
26	1.625	2.527	2.666	2.809	2.934	3.083	3.271
27	1.688	2.590	2.729	2.872	2.997	3.146	3.334
28	1.750	2.652	2.791	2.934	3.059	3.208	3.396
29	1.812	2.714	2.853	2.996	3.121	3.270	3.458
30	1.875	2.777	2.916	3.059	3.184	3.333	3.521
31	1.938	2.840	2.979	3.122	3.247	3.396	3.584
32	2.000	2.902	3.041	3.184	3.309	3.458	3.646
34	2.125	3.027	3.166	3.309	3.434	3.583	3.771
36	2.250	3.152	3.291	3.434	3.559	3.708	3.896
38	2.375	3.277	3.416	3.559	3.684	3.833	4.021
40	2.500	3.402	3.541	3.684	3.809	3.958	4.146
42	2.625	3.527	3.666	3.809	3.934	4.083	4.271
44	2.750	3.652	3.791	3.934	4.059	4.208	4.396
46	2.875	3.777	3.916	4.059	4.184	4.333	4.521
48	3.000	3.902	4.041	4.184	4.309	4.458	4.646
50	3.125	4.027	4.166	4.309	4.434	4.583	4.771
52	3.250	4.152	4.291	4.434	4.559	4.708	4.896
54	3.375	4.277	4.416	4.559	4.684	4.833	5.021
56	3.500	4.402	4.541	4.684	4.809	4.958	5.146
58	3.625	4.527	4.666	4.809	4.934	5.083	5.271
60	3.750	4.652	4.791	4.934	5.059	5.208	5.396
62	3.875	4.777	4.916	5.059	5.184	5.333	5.521
64	4.000	4.902	5.041	5.184	5.309	5.458	5.646

SEE SHEET 5 REGARDING INTERMEDIATE OR LONGER LENGTHS.

② COMPLETELY REVISED

NAS6703 THRU 6720 ②
SHEET 6

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.0156 AND .0312 OVERSIZE SHANK FOR REPLACEMENT OF BOLTS SHOWN ON SHEET 1

Head marking: Same as shown on sheet 1
plus identification for oversize, as applicable,
to be included in second sector.
Identify .0156 oversize by "X".
Identify .0312 oversize by "Y".

See Sheets 1 thru 6 for dimensions not shown.

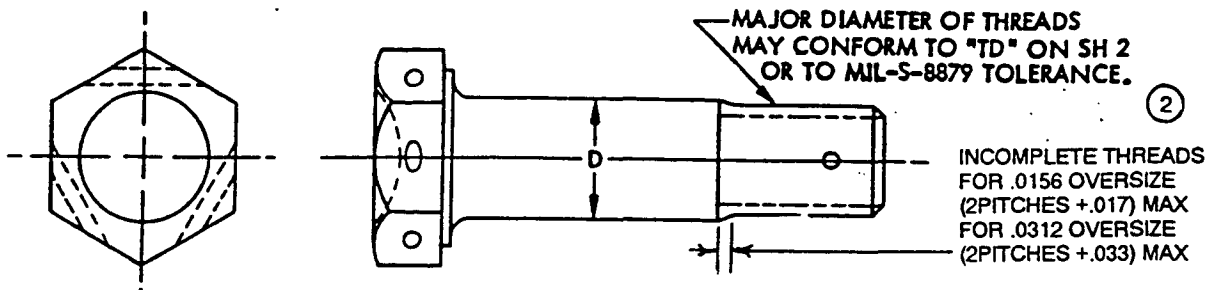


TABLE III OVERSIZE PART NUMBERS AND DIMENSIONS

PART NUMBER, **CAD PLATED, NONLOCKING, ***UNDRILLED HEAD, ****UNDRILLED SHANK	NOMINAL THREAD SIZE	D DIA. .0156 OVERSIZE SHANK (†) (2)							
		UNPLATED BOLTS		PLATED OR COATED BOLTS					
				BEFORE CAD PLATE		BEFORE AL COATING		AFTER PLATING -OR- COATING-	
		MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN
NAS6703-X	.1900-32	.2026	.2021	.2016	.2010	.2018	.2012	.2026	.2016
NAS6704-X	.2500-28	.2651	.2646	.2641	.2635	.2643	.2637	.2651	.2641
NAS6705-X	.3125-24	.3276	.3271	.3266	.3260	.3268	.3262	.3276	.3266
NAS6706-X	.3750-24	.3901	.3896	.3891	.3885	.3893	.3887	.3901	.3891
NAS6707-X	.4375-20	.4526	.4521	.4516	.4510	.4518	.4512	.4526	.4516
NAS6708-X	.5000-20	.5151	.5146	.5141	.5135	.5143	.5137	.5151	.5141
NAS6709-X	.5625-18	.5771	.5766	.5761	.5755	.5763	.5757	.5771	.5761
NAS6710-X	.6250-18	.6396	.6391	.6386	.6380	.6388	.6382	.6396	.6386
NAS6712-X	.7500-16	.7646	.7641	.7636	.7630	.7638	.7632	.7646	.7636
NAS6714-X	.8750-14	.8896	.8891	.8886	.8880	.8888	.8882	.8896	.8886
NAS6716-X	1.0000-12	1.0146	1.0141	1.0136	1.0130	1.0138	1.0132	1.0146	1.0136
NAS6718-X	1.1250-12	1.1396	1.1386	1.1386	1.1375	1.1388	1.1377	1.1396	1.1381
NAS6720-X	1.2500-12	1.2646	1.2636	1.2636	1.2625	1.2638	1.2627	1.2646	1.2631
D DIA. .0312 OVERSIZE SHANK (†)									
NAS6703-Y	.1900-32	.2182	.2177	.2172	.2166	.2174	.2168	.2182	.2172
NAS6704-Y	.2500-28	.2807	.2802	.2797	.2791	.2799	.2793	.2807	.2797
NAS6705-Y	.3125-24	.3432	.3427	.3422	.3416	.3424	.3418	.3432	.3422
NAS6706-Y	.3750-24	.4057	.4052	.4047	.4041	.4049	.4043	.4057	.4047
NAS6707-Y	.4375-20	.4682	.4677	.4672	.4666	.4674	.4668	.4682	.4672
NAS6708-Y	.5000-20	.5307	.5302	.5297	.5291	.5299	.5293	.5307	.5297
NAS6709-Y	.5625-18	.5927	.5922	.5917	.5911	.5919	.5913	.5927	.5917
NAS6710-Y	.6250-18	.6552	.6547	.6542	.6536	.6544	.6538	.6552	.6542
NAS6712-Y	.7500-16	.7802	.7797	.7792	.7786	.7794	.7788	.7802	.7792
NAS6714-Y	.8750-14	.9052	.9047	.9042	.9036	.9044	.9038	.9052	.9042
NAS6716-Y	1.0000-12	1.0302	1.0297	1.0292	1.0286	1.0294	1.0288	1.0302	1.0292
NAS6718-Y	1.1250-12	1.1552	1.1542	1.1542	1.1531	1.1544	1.1533	1.1552	1.1537
NAS6720-Y	1.2500-12	1.2802	1.2792	1.2792	1.2781	1.2794	1.2783	1.2802	1.2787

* Grip dash number in .0625 increments. See sheets 5 and 6 for grip and length dimensions. For material, finish and procurement information, see sheets 3 & 4.

** For unplated bolt, add "U" after basic number.

EXAMPLE: NAS6704HU-X = Drilled head, unplated, .0156 oversize bolt.

For aluminum coated bolt, add "A" after basic number.

EXAMPLE: NAS6704A-X = Undrilled aluminum coated, .0156 oversize bolt.

*** For drilled head, add "H" after basic number.

EXAMPLE: NAS6704H-X = Drilled head, Cad plated, .0156 oversize bolt.

**** For drilled shank, add "D" after basic number.

EXAMPLE: NAS6704D-X = Drilled shank, Cad plated, .0156 oversize bolt.

NAS6703 THRU 6720
SHEET 7

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